Pre-Basque Derivational Morphology: A Minimal Reconstruction

Steve Girard

Independent Researcher

K.

AI Research Collaborator

May 2026

Abstract

This paper reconstructs the core derivational mechanisms of Pre-Basque (henceforth Pre-B), an internally coherent morphological layer anterior to both Proto-Basque and the documented stages of Basque. Building on the operational framework established in Girard (2026a), the study isolates six derivational patterns that satisfy the criteria of internal recurrence, independent segmentability, and co-distribution with other Pre-B traits. These patterns—localization (-(V)r/-(V)ra), collectivization (-ta/-eta), individuation (-na), light nominalization (a-/e-), agentive/instrumental derivation (-ri), and qualitative specification (s $\sim$ z)—form a structured and semantically transparent system irreducible to modern Basque or Proto-Basque morphology. Their interactions reveal a layered derivational architecture in which morphological operations combine in predictable sequences. The minimal model proposed here provides a conservative yet testable representation of this architecture, demonstrating that Pre-B constituted a morphologically active system governed by its own derivational grammar. These findings refine the internal stratification of the Basque lexicon and lay a foundation for future work on Pre-B phonotactics, lexical structure, and the emergence of later morphosyntactic categories.

Keywords: *Pre-Basque, derivational morphology, lexical stratification, internal reconstruction, morphological architecture*

1. Introduction

The study of Pre-Basque (Pre-B) morphology has so far been devoted primarily to segmentation criteria, lexical stratification, and the reconstruction of the deictic system. While these lines of inquiry have established the internal coherence and autonomy of the Pre-B layer, the structure of its derivational morphology remains largely unexplored. Yet derivation constitutes a crucial component of any morphological system: it reveals how lexical bases were shaped, extended, and functionally specified within the language.

Building on the operational definition of Pre-B established in Girard (2026a), this paper undertakes a systematic examination of the internal derivational mechanisms that can be reconstructed without recourse to Proto-Basque or modern Basque. The goal is not to project later morphological categories backward but rather to identify derivational patterns that are segmentable, recurrent, and co-distributed with other Pre-B traits. This approach ensures that the resulting model remains falsifiable, reproducible, and independent of subsequent developments in the language.

The analysis identifies six core derivational patterns satisfying these criteria: (i) -(V)r/-(V)ra, forming localized or concrete entities; (ii) -ta/-eta, forming collectives or resultative states; (iii) -na, marking individuated or delimited entities; (iv) the vocalic prefix a-/e-, functioning as a light nominalizer; (v) -ri, deriving minimal agentive or instrumental nouns; and (vi) the s $\sim$ z alternation, expressing qualitative specification. These patterns are no longer productive in modern Basque and cannot be derived from Proto-Basque morphology, indicating that they belong to an earlier morphological stratum.

Beyond the identification of individual patterns, the paper demonstrates that these mechanisms interact in systematic ways, revealing a layered derivational architecture within Pre-B. Certain patterns combine in predictable sequences (e.g., localization $\rightarrow$ collectivization $\rightarrow$ individuation), while others exhibit functional complementarity or domain-specific constraints. These interactions furnish independent evidence for the internal coherence of the Pre-B morphological system and for the viability of its reconstruction on strictly internal grounds.

2. Theoretical and methodological framework

This study follows the methodological principles established in Girard (2026a), where Pre-Basque (Pre-B) is defined as an internally coherent morpho-lexical stratum anterior to both Proto-Basque and the documented stages of Basque. The present analysis applies this framework specifically to derivational morphology, drawing on the three core criteria of internal recurrence, independent segmentability, and co-distribution with other Pre-B traits. No Proto-Basque reconstructions, no modern Basque categories, and no external comparanda are invoked at any stage. All patterns are reconstructed exclusively on the basis of internal evidence, using archaic lexical bases and opaque forms as the primary data source.

A derivational mechanism is attributed to the Pre-B layer only if it satisfies three independent criteria:

> (i) **Internal recurrence.** The pattern must appear across multiple lexemes exhibiting comparable structure and function. Isolated formations are excluded.
>
> (ii) **Independent segmentability.** The morpheme must be segmentable without reference to Proto-Basque morphology, modern Basque derivation, or later analogical developments. Segmentations that presuppose modern categories (e.g., adjectives, articles, productive suffixes) are rejected.
>
> (iii) **Co-distribution with other Pre-B traits.** The pattern must co-occur with phonotactic, morphological, or lexical properties already identified as Pre-B (e.g., archaic alternations, non-modern functions, opaque lexemes). This requirement ensures that the derivational mechanism belongs to the same structural layer.

Conversely, a derivational pattern is excluded from the Pre-B layer if its function matches a productive modern Basque morpheme, its distribution corresponds to modern categories (e.g., adjectival *-ko*), its morphophonology reflects later regularizations, or its segmentation depends on Proto-Basque reconstructions. Independent onomastic evidence further supports this exclusion logic: Faria (2025: 156) demonstrates that certain short Paleo-Basque lexical items chronologically precede their longer counterparts in the epigraphic record, confirming that abbreviated forms are not late derivations and that the antiquity of monosyllabic bases can be independently verified.

The goal, then, is not to project modern morphology backward, but to identify derivational mechanisms that are internally reconstructible within the Pre-B lexicon. The justification for extending internal reconstruction to derivation rests on the observation that derivational morphology is inherently combinatorial: if a stratum possesses recurrent and segmentable formatives that co-occur systematically, the derivational system itself becomes a legitimate object of reconstruction. This approach ensures that the resulting model is falsifiable, reproducible, and independent of later Basque developments. All examples presented in this paper have been segmented using the operational criteria defined in Girard (2026a). The analysis is strictly internal: no external etymologies, no proto-forms, and no modern analogies are invoked.

3. Data and corpus

The analysis presented in this paper is based exclusively on lexical material satisfying the operational criteria for Pre-B reconstruction defined in Girard (2026a). The corpus consists of archaic lexical bases and derived forms whose segmentation, distribution, and semantic profiles are internally coherent and do not depend on Proto-Basque or modern Basque morphology. No external etymologies, proto-forms, or modern analogies are used at any stage.

3.1. Nature of the data

The corpus includes: (i) archaic lexical bases (e.g., *har*, *lur*, *zur*, *bel*, *suk*, *bas*, *las*, *es*); (ii) derived forms exhibiting recurrent Pre-B patterns (e.g., *har-a*, *lur-na*, *zur-ta*, *har-ri*, *baz-*); (iii) morphologically transparent alternations (e.g., s $\sim$ z); and (iv) forms showing co-distribution with other Pre-B traits (e.g., localization + collectivization + individuation).

All items included in the corpus meet the following conditions: (a) internal segmentability without reference to Proto-Basque or modern Basque; (b) recurrent morphological behavior across multiple bases; (c) compatibility with other Pre-B features, including phonotactic and deictic properties; and (d) absence of modern morphological influence, analogical reshaping, or later regularization.

3.2. Criteria for inclusion

A form is included in the corpus only if its segmentation is independent of modern categories (e.g., no reliance on the modern article, adjectival morphology, or case markers), its derivational structure is internally reproducible, its semantic shift is consistent with Pre-B derivational functions, and its morphology is coherent with other Pre-B patterns. Forms requiring Proto-Basque reconstruction, modern analogies, or external etymologies are excluded.

3.3. Criteria for exclusion

The following types of data are systematically excluded: modern Basque derivations (e.g., *-ko*, *-tasun*, *-keria*), forms whose segmentation depends on the modern article *-a*, Proto-Basque reconstructions or proto-forms, analogical formations from later stages of the language, and items whose morphology is opaque solely because of later sound change. This ensures that the corpus reflects Pre-B morphology alone, not subsequent developments.

3.4. Corpus structure

The corpus is organized around lexical bases, derived forms, morphological alternations, derivational sequences (e.g., *har* → *har-a* → *har-ta* → *har-na*), and compatibility patterns across suffixes and alternations. This structure facilitates the identification of recurrent derivational mechanisms and layered morphological behavior, which form the basis of the analyses in Section 5 and the architectural model in Section 7.

3.5. Relation to previous work

The corpus is consistent with the segmentation principles established in Girard (2026a; 2026b), the Pre-B deictic system reconstructed in Girard & K. (2026d), and the internal stratification criteria employed in earlier studies of Pre-B morphology (Girard 2026b; 2026c). However, the present corpus is independent of those works: it is constructed solely on the basis of derivational evidence and does not rely on deictic or phonotactic criteria except where co-distribution is relevant.

4. Overview of Pre-Basque derivational morphology

Pre-Basque derivational morphology exhibits a coherent set of mechanisms that shape lexical bases into more concrete, collective, individuated, or functionally specified forms. Unlike modern Basque, where derivation interacts with syntactic categories and productive morphological classes, Pre-B derivation operates through a small number of semantically stable and internally consistent processes. These processes are not recoverable from Proto-Basque and do not correspond to modern categories, confirming that they belong to an earlier morphological stratum. Crucially, the patterns identified here are characterized by light morphology, semantic transparency, restricted combinatorics, and full internal autonomy—properties consistent with the stratification proposed in Girard (2026a).

4.1. General characteristics of the system

The Pre-B derivational system is characterized by: (i) light morphology—short, segmentable suffixes and prefixes; (ii) semantic transparency—each pattern contributes a clear and consistent semantic operation; (iii) restricted combinatorics—patterns combine in predictable sequences rather than freely; (iv) absence of modern categories—no adjectival morphology, no article-based derivation, no productive case-driven processes; and (v) internal autonomy—derivation is reconstructible without reference to Proto-Basque or modern Basque. These characteristics point to a system that is lexical, morphological, and pre-syntactic.

4.2. Types of derivational operations

Pre-B derivation operates through a limited set of semantic transformations:

No.	Pattern	Function
(i)	-(V)r / -(V)ra	Localization and concreteness — derives materially bounded or spatially delimited entities
(ii)	-ta / -eta	Collectivization and resultative formation — derives aggregates, masses, or states resulting from a base
(iii)	-na	Individuation and referential delimitation — derives defined or individuated entities
(iv)	a- / e-	Light nominalization and morphological shaping — stabilizes or nominalizes lexical bases
(v)	-ri	Agentive and instrumental characterization — derives minimal agents or instruments
(vi)	s $\sim$ z	Qualitative specification — derives intensified or specified qualitative forms

These operations are semantically complementary and morphologically stable, forming the backbone of Pre-B derivation.

4.3. Distribution across lexical bases

The derivational patterns are attested across a set of archaic lexical bases, many of which also participate in the Pre-B deictic system. Typical bases include *har* "stone," *lur* "earth," *zur* "wood," *bel* "black," *suk* "heat," and *bas* / *las* / *es* (qualitative bases).

The recurrence of derivation across these bases indicates that the system was productive, not merely lexicalized.

4.4. Morphophonological behavior

Pre-B derivation exhibits several morphophonological properties: stable suffixation (*-r*, *-ra*, *-ta*, *-na*, *-ri*), light prefixation (*a-* / *e-*), non-phonological alternation (s $\sim$ z), and absence of epenthetic or analogical reshaping characteristic of later Basque. The morphophonology is minimal but systematic, reinforcing the internal coherence of the system.

4.5. Absence of modern and Proto-Basque categories

Pre-B derivation does not rely on modern adjectival morphology, the modern article, Proto-Basque case morphology, modern derivational suffixes (e.g., *-ko*, *-tasun*, *-keria*), or any syntactic categories. This absence is not a gap but a diagnostic feature: it confirms that Pre-B derivation belongs to a morphological layer prior to the emergence of these categories.

4.6. Toward a structured derivational architecture

The patterns identified here do not form a flat inventory. Their distribution and compatibility (developed in Section 6) reveal layering (localization $\rightarrow$ collectivization $\rightarrow$ individuation), functional complementarity, domain-specific constraints, and cross-pattern reinforcement across the same bases. These properties justify the reconstruction of a minimal derivational architecture, presented in Section 7.

5. Core derivational patterns

5.1. The pattern -(V)r / -(V)ra: localization and concreteness

The suffixal pattern -(V)r / -(V)ra is one of the most stable and internally coherent derivational mechanisms identifiable within the Pre-B lexicon. It forms nouns denoting localized, concrete, or materially delimited entities. Its function, distribution, and morphophonological behavior differ systematically from the corresponding formations in modern Basque, and its segmentation does not rely on Proto-Basque morphology. The pattern therefore satisfies all three criteria for Pre-B reconstruction.

5.1.1. Form and function

The pattern consists of a base (typically monosyllabic or disyllabic) followed by a rhotic element *-r* or its extended form *-ra*. The resulting formation denotes a materially bounded entity, often the concrete instantiation of a more abstract or undifferentiated base. This function is not equivalent to any productive derivational category in modern Basque, where *-ra* is primarily directional and *-r* is not a nominalizer. In Pre-B, the suffix contributes localization, concreteness, or material delimitation, without directional meaning.

5.1.2. Examples

The pattern is recurrent across several of the most archaic lexical bases:

har → *har-a* “stone → stone-mass, concretized stone entity”
lur → *lur-a* “earth → ground, delimited portion of earth”
zur → *zur-a* “wood → wood-matter, concrete wooden entity”

These formations exhibit a consistent semantic shift from material substance to concrete instantiation, a shift not paralleled in the modern language.

5.1.3. Arguments for Pre-B status

(i) *Functional incompatibility with modern Basque:* modern Basque does not use *-r* or *-ra* as nominalizers of material entities. (ii) *Morphophonological stability:* the rhotic element appears with remarkable regularity across unrelated bases, suggesting a productive mechanism rather than isolated lexical fossilization. (iii) *Internal recurrence:* the pattern is attested across multiple lexemes with comparable semantic effects. (iv) *Independent segmentability:* the segmentation base + -(V)r(a) is possible without reference to Proto-Basque reconstructions or modern categories. (v) *Co-distribution with other Pre-B traits:* the bases involved (*har*, *lur*, *zur*) also participate in other Pre-B derivational patterns (e.g., *-ta*, *-na*), indicating that -(V)r(a) belongs to the same structural layer.

5.1.4. Implications

The pattern suggests that Pre-B possessed a productive mechanism for deriving concrete, materially bounded nouns from more abstract bases. This mechanism is absent from the modern language and cannot be reconstructed from Proto-Basque, indicating that it belongs to an earlier morphological stratum. Its interaction with other Pre-B suffixes (e.g., *har-a* → *har-ta*, *lur-a* → *lur-na*) further supports the existence of a structured derivational architecture within Pre-B.

5.2. The pattern -ta / -eta: collectives and resultative states

The suffixal pattern -ta / -eta constitutes one of the most robust derivational mechanisms identifiable within the Pre-B lexicon. It forms nouns denoting collective entities, material aggregates, or resultative states. Its semantic range, morphophonological behavior, and distribution differ fundamentally from the functions associated with *-ta* in modern Basque, where the suffix is largely restricted to abstract nouns or lexicalized formations. In Pre-B, the pattern is productive, semantically transparent, and structurally coherent.

5.2.1. Form and function

The pattern consists of a base followed by *-ta* or its extended variant *-eta*. The resulting formation denotes either (a) a collective or mass entity derived from a material base, or (b) a resultative state derived from a qualitative or material base. These functions are not aligned with the modern language, where *-ta* is not a productive collective marker and does not systematically derive material aggregates.

5.2.2. Examples

har → *har-ta* "stone → stone-mass, stony aggregate"
zur → *zur-ta* "wood → mass of wood, wooden aggregate"
bel → *bel-ta* "black → blackness, state of being black"

These formations exhibit a consistent semantic shift from material or qualitative base to collective or resultative entity, a shift that is not productive in modern Basque and cannot be derived from Proto-Basque morphology.

5.2.3. Arguments for Pre-B status

(i) *Functional incompatibility with modern Basque:* modern Basque does not use *-ta* to derive collectives from material bases. (ii) *Internal recurrence:* the pattern appears across multiple bases with comparable semantic effects. (iii) *Independent segmentability:*

the segmentation base + *-ta*/*-eta* is possible without reference to Proto-Basque reconstructions or modern derivational categories. (iv) *Co-distribution with other Pre-B traits:* the bases involved (*har*, *zur*, *bel*) also participate in other Pre-B derivational patterns. (v) *Morphological stability:* the suffix appears with remarkable regularity and semantic coherence.

5.2.4. Implications

The pattern indicates that Pre-B possessed a productive mechanism for deriving collective and resultative nouns from material or qualitative bases. Its interaction with other Pre-B suffixes (e.g., *har-a* → *har-ta*, *bel* → *bel-ta*) further supports the existence of a structured derivational architecture within Pre-B.

5.3. The pattern -na: defined and individuated entities

The suffix *-na* constitutes one of the clearest derivational mechanisms within the Pre-B lexicon. It forms nouns denoting defined, individuated, or delimited entities. Its function is neither equivalent to the modern Basque article nor to any productive nominal suffix in the contemporary language. Instead, *-na* reflects an earlier morphological strategy for marking referential delimitation within the Pre-B system.

5.3.1. Form and function

The pattern consists of a lexical base followed by *-na* (or its reduced variant *-n*). The resulting formation denotes a defined entity, a delimited portion of a substance, or a referentially individuated instance of the base. This function is not equivalent to the modern Basque article *-a*, nor to any modern derivational category. In Pre-B, *-na* does not encode definiteness in the modern sense; rather, it marks referential delimitation internal to the lexicon, consistent with the deictic architecture reconstructed in Girard (2026a).

5.3.2. Examples

lur → *lur-na* "earth → the earth (as a delimited portion), defined ground entity"
ur → *ur-na* "water → the water (as a bounded quantity), individuated liquid entity"
har → *har-na* "stone → the stone (as a specific, delimited stone entity)"

These formations exhibit a consistent semantic shift from substance or material to defined, individuated entity.

5.3.3. Arguments for Pre-B status

(i) *Functional incompatibility with modern Basque:* modern Basque uses *-a* as an article, not as a derivational marker of referential delimitation. The Pre-B *-na* pattern is therefore not a precursor of the modern article and cannot be explained through modern categories. (ii) *Internal recurrence.* (iii) *Independent segmentability* without reference to Proto-Basque reconstructions or modern definiteness marking. (iv) *Co-distribution with other Pre-B traits.* (v) *Compatibility with the Pre-B deictic system* (Girard 2026a).

5.3.4. Implications

The pattern suggests that Pre-B possessed a morphological strategy for marking referential delimitation distinct from both Proto-Basque and modern Basque. Its interaction with other Pre-B suffixes (e.g., *lur* → *lur-a* → *lur-na*) supports the existence of a layered derivational architecture in which localization and individuation operate in complementary ways.

5.4. The vocalic prefix a- / e-: light nominalization

The vocalic prefix a- / e- constitutes a light but recurrent derivational mechanism within the Pre-B lexicon. It forms nominalized or morphologically "shaped" entities without any connection to the modern Basque article or to later definiteness marking. In Pre-B, the prefix does not encode deixis, definiteness, or case; instead, it functions as a minimal morphological operator that derives lexical nouns from more abstract bases or restructures the phonological profile of the base.

5.4.1. Form and function

The pattern consists of a simple vocalic prefix (*a-* or *e-*) attached to a lexical base. Its functions include: (a) light nominalization; (b) morphological shaping of the base; (c) creation of a more lexicalized or referentially stable form; and (d) phonotactic stabilization in forms where the base begins with a consonant cluster or unstable onset. Crucially, this prefix is not the modern Basque article *-a*, nor a precursor of it. Its function is derivational, not syntactic or deictic.

5.4.2. Examples

a-lur "earth → (the) earth as a lexicalized noun; nominalized ground entity"
e-har "stone → (the) stone as a lexicalized form; morphologically shaped stone entity"
a-zur "wood → (the) wood as a stable lexical noun; nominalized wooden entity"

5.4.3. Arguments for Pre-B status

(i) *Functional incompatibility with modern Basque:* modern Basque does not use *a-* or *e-* as derivational prefixes. (ii) *Internal recurrence* across multiple bases. (iii) *Independent segmentability.* (iv) *Co-distribution with other Pre-B traits.* (v) *Morphological stability and phonotactic motivation:* the prefix often appears in contexts where the base begins with a consonant cluster or unstable onset, suggesting a phonotactically motivated but morphologically productive mechanism.

5.4.4. Implications

The pattern indicates that Pre-B possessed a light, productive nominalization strategy distinct from both Proto-Basque and modern Basque. Its interaction with other Pre-B suffixes (e.g., *a-lur* → *lur-a* → *lur-na*) supports the existence of a layered derivational architecture in which prefixation and suffixation operate in complementary ways.

5.5. The pattern -ri: minimal agentive and instrumental derivation

The suffix *-ri* constitutes a discrete but highly diagnostic derivational mechanism within the Pre-B lexicon. It forms nouns denoting minimal agents, instruments, or entities characterized by a functional relation to the base. Its semantic range and distribution differ fundamentally from the uses of *-ri* in modern Basque, where the suffix is not a productive agentive or instrumental marker.

5.5.1. Form and function

The pattern consists of a lexical base followed by *-ri* (or its reduced variant *-r*). The resulting formation denotes: (a) a minimal agent ("that which performs or enacts X");

(b) a minimal instrument ("that which serves for X"); or (c) a functionally characterized entity derived from a material or qualitative base.

5.5.2. Examples

suk → *suk-ri* "heat → that which heats; thermal agent"
har → *har-ri* "stone → stony implement; entity functioning as stone"
zur → *zur-ri* "wood → wooden tool; entity functioning as wood"

5.5.3. Arguments for Pre-B status

(i) *Functional incompatibility with modern Basque:* modern Basque does not use *-ri* as an agentive or instrumental suffix. (ii) *Internal recurrence* across multiple bases. (iii) *Independent segmentability.* (iv) *Co-distribution with other Pre-B traits.* (v) *Morphological stability.*

5.5.4. Implications

The pattern indicates that Pre-B possessed a productive mechanism for deriving agentive and instrumental nouns. Its interaction with other Pre-B suffixes (e.g., *har* → *har-ri* → *har-ta*) supports a structured architecture in which functional characterization (*-ri*) and material or collective derivation (*-ta*, -(V)r(a)) operate in complementary ways.

5.6. The s ∼ z alternation: qualitative specification

The alternation s ∼ z constitutes a morphologically driven derivational mechanism within the Pre-B lexicon. It produces forms with a qualitative, intensifying, or specifying function. Crucially, this alternation is not phonological in nature: it does not correspond to any regular sound change in modern Basque, nor to any Proto-Basque reconstruction. Instead, it reflects a productive morphological strategy internal to the Pre-B system.

5.6.1. Form and function

The pattern consists of a base containing *s* that alternates with a derived form containing *z*. The resulting formation typically expresses: (a) qualitative specification; (b) light intensification; or (c) a derived or modified state relative to the base. This alternation is not phonologically conditioned and does not correspond to any known historical sound law. In Pre-B, it functions as a morphological operator, not as a phonetic process.

5.6.2. Examples

bas → *baz-* "low / base → specified or intensified form"
las → *laz-* "loose / slack → derived qualitative form"
es → *ez-* "raw / unformed → derived or intensified qualitative state"

5.6.3. Arguments for Pre-B status

(i) *Non-phonological nature of the alternation:* the alternation s → z is not conditioned by phonological environment and does not correspond to any modern or Proto-Basque sound change. (ii) *Functional incompatibility with modern Basque.* (iii) *Internal recurrence.* (iv) *Independent segmentability.* (v) *Co-distribution with other Pre-B traits.*

5.6.4. Implications

The pattern indicates that Pre-B possessed a morphological alternation system capable of deriving qualitative or intensified forms through consonantal modification. Its interaction with other Pre-B derivational processes (e.g., *bas* → *baz-* → *baz-ta*) supports a structured and multi-layered derivational architecture.

6. Interactions between patterns

The six derivational patterns identified in this study do not operate in isolation. Their co-distribution, compatibility constraints, and sequential behavior reveal a structured derivational architecture within the Pre-B lexicon. This architecture is not recoverable from modern Basque and cannot be derived from Proto-Basque morphology, indicating that the patterns belong to an earlier morphological stratum. The interactions documented below provide independent evidence for the internal coherence of the Pre-B system and constitute a key argument for treating Pre-B derivation as a unified grammatical domain rather than a flat inventory of fossilized suffixes.

6.1. Sequential compatibility and layering

Several patterns exhibit ordered compatibility, suggesting that Pre-B derivation proceeded through identifiable morphological layers. The most common sequence is:

(i) base → localization / concreteness (-(V)r / -(V)ra)
(ii) localized entity → collective or resultative (-ta / -eta)
(iii) collective or localized entity → individuated form (-na)

For example:

har → har-a → har-ta → har-na
"stone → stone-entity → stony mass → the defined stone entity"

This sequence is not productive in modern Basque and has no Proto-Basque analogue. It reflects a Pre-B architecture in which localization precedes collectivization, and individuation applies to already-derived forms.

6.2. Complementarity of functions

The patterns exhibit functional complementarity, each contributing a distinct semantic operation:

Pattern	Semantic contribution
-(V)r(a)	Localization and concreteness
-ta / -eta	Collectivization and resultative state
-na	Individuation and referential delimitation
a- / e-	Nominalization and morphological shaping
-ri	Agentive / instrumental characterization
$s \sim z$	Qualitative specification / intensification

These functions do not overlap in the modern language, nor do they correspond to Proto-Basque categories. Their complementarity supports the hypothesis of a coherent Pre-B derivational system.

6.3. Cross-pattern reinforcement

Certain bases participate in multiple patterns, creating reinforcement chains that reveal the internal logic of Pre-B morphology:

Base	Attested derived forms
lur	*a-lur, lur-a, lur-ta, lur-na*
har	*har-a, har-ta, har-ri, har-na*
zur	*a-zur, zur-a, zur-ta, zur-ri*

This cross-pattern participation demonstrates that the patterns are not accidental but form part of a shared morphological system.

6.4. Mutual exclusivity and constraints

Some patterns exhibit mutual exclusivity, indicating that Pre-B derivation was governed by structural constraints rather than free combination. Specifically, *-ri* (agentive/instrumental) rarely co-occurs with *-na* (individuation), suggesting that functional characterization and referential delimitation were distinct morphological operations. Similarly, the s $\sim$ z alternation typically applies to qualitative bases rather than material ones, indicating a domain-specific morphological rule. These constraints reinforce the idea of a structured derivational grammar.

6.5. Implications for Pre-B morphological architecture

The interactions documented above support several conclusions: (i) Pre-B derivation was layered, with identifiable sequences of morphological operations; (ii) patterns were semantically specialized, each contributing a distinct type of lexical transformation; (iii) patterns interacted systematically, revealing a coherent internal architecture; (iv) the system is not recoverable from modern Basque, confirming its Pre-B status; and (v) the architecture is compatible with the Pre-B deictic system reconstructed in Girard (2026a), suggesting that derivation and deixis formed an integrated morphological domain.

7. A minimal model of Pre-Basque derivational architecture

The six derivational patterns identified in this study reveal a coherent and internally structured morphological system within Pre-B. Although the available data do not permit a full reconstruction of the entire derivational grammar, they are sufficient to propose a minimal model capturing the core mechanisms, their functional domains, and their sequential interactions. This model is intentionally conservative: it includes only those components that are recurrent, segmentable, and co-distributed with other Pre-B traits, in accordance with the methodological criteria defined in Girard (2026a).

7.1. Core components of the system

The Pre-B derivational system can be represented as a set of six productive mechanisms, each contributing a distinct type of lexical transformation:

Each mechanism operates independently but interacts with others in predictable ways, forming a structured derivational architecture.

Mechanism	Form	Function
Localization / concreteness	-(V)r / -(V)ra	Derives materially bounded entities
Collectivization / resultative	-ta / -eta	Derives aggregates or resultative states
Individuation / delimitation	-na	Derives defined or individuated entities
Light nominalization	a- / e-	Stabilizes or nominalizes bases
Agentive / instrumental	-ri	Derives agents or instruments
Qualitative specification	s $\sim$ z	Derives intensified qualitative forms

7.2. Layered derivational structure

The interactions documented in Section 6 indicate that Pre-B derivation proceeded through ordered morphological layers rather than through free combination. The minimal architecture can be represented as follows:

Base level: Raw lexical bases (often monosyllabic or disyllabic).

Level I — Light morphological shaping: a- / e- (optional). Provides nominalization or phonotactic stabilization.

Level II — Material or spatial specification: -(V)r / -(V)ra. Derives localized or concrete entities.

Level III — Collective or resultative formation: -ta / -eta. Applies to bases or to Level II derivatives.

Level IV — Individuation: -na. Applies to bases or to Level II/III derivatives.

Parallel qualitative modification: s $\sim$ z alternation. Applies to qualitative bases at any stage prior to suffixation.

This layered structure is not recoverable from modern Basque and has no Proto-Basque analogue, indicating that it reflects an earlier morphological system.

7.3. Compatibility and constraints

The model incorporates several compatibility constraints observed in the data: *-ri* (agentive/instrumental) typically applies to Level I or Level II forms but rarely to individuated forms (*-na*); the s $\sim$ z alternation applies primarily to qualitative bases, not to material ones; *-na* (individuation) tends to apply after localization or collectivization, but not after agentive derivation (*-ri*); and a-/e- is compatible with all suffixes but does not itself trigger further derivation.

7.4. Minimal schematic representation

The minimal model can be summarized schematically:

Primary derivational chain:

BASE → (a- / e-) → (-(V)r / -(V)ra) → (-ta / -eta) → (-na)

Parallel optional modification:

BASE → BASE(*z*) (qualitative specification)

Functional derivation:

BASE → BASE-*ri* (agentive / instrumental)

This schematic captures the essential architecture without over-interpreting the data.

7.5. Implications for Pre-B reconstruction

The minimal model has several implications: (i) Pre-B possessed a multi-layered derivational system, not a flat or unstructured morphology; (ii) derivation preceded lexicalization, indicating an active morphological grammar; (iii) the system is autonomous, with no dependence on modern Basque or Proto-Basque categories; (iv) the architecture aligns with the Pre-B deictic system, suggesting that deixis and derivation formed an integrated morphological domain; and (v) the model provides a testable framework for future work on Pre-B phonotactics, lexical stratification, and morphosyntax.

8. Implications for Pre-Basque reconstruction

The derivational architecture reconstructed in this study has significant implications for our understanding of Pre-B morphology, lexical structure, and historical development. The six patterns identified here—together with their interactions and layered organization—provide independent evidence for the existence of a coherent morphological system that predates both Proto-Basque and the documented stages of Basque. This section outlines the principal consequences of these findings.

8.1. A morphologically active Pre-Basque layer

The presence of productive derivational mechanisms indicates that Pre-B was not a purely lexical or fossilized stratum. Instead, it possessed active morphological processes, systematic semantic operations, predictable combinatorics, and a structured derivational grammar. This finding challenges the traditional view of pre-Basque as a static lexical substrate and supports the hypothesis of a fully functional morphological system.

8.2. Independence from Proto-Basque and modern Basque

None of the six derivational patterns corresponds to productive morphology in modern Basque, nor can they be derived from Proto-Basque reconstructions. This independence has two major implications: first, Pre-B morphology is not a proto-stage of modern Basque morphology but represents a distinct structural layer with its own categories and operations; second, Proto-Basque cannot serve as a basis for reconstructing Pre-B derivation—the direction of inference must be strictly internal, as defined in Girard (2026a).

8.3. Integration with the Pre-B deictic system

The derivational patterns—especially *-na* (individuation) and -(V)r(a) (localization)—align closely with the Pre-B deictic architecture reconstructed in Girard (2026a). This suggests that deixis and derivation were not separate domains, that referential delimitation was partly morphological, and that Pre-B possessed an integrated system for marking localization, collectivization, and individuation.

8.4. Implications for lexical stratification

The derivational architecture provides new tools for distinguishing Pre-B lexical items from later Proto-Basque formations and modern analogical developments. Bases participating in multiple Pre-B patterns (e.g., *har*, *lur*, *zur*) can be securely assigned to the Pre-B stratum; derived forms such as *har-ta*, *lur-na*, and *zur-ri* cannot be explained by later morphology; and qualitative alternations (s $\sim$ z) provide a diagnostic for identifying archaic qualitative bases. This allows for a more precise stratification of the Basque lexicon.

8.5. Implications for phonotactics and morphophonology

The derivational patterns shed light on Pre-B phonotactics: the vocalic prefix a-/e- suggests constraints on initial clusters; the stability of *-r*, *-t*, *-n* points to early consonantal morphology; and the non-phonological s $\sim$ z alternation indicates a morphological alternation system independent of sound change. These observations open the way for a future reconstruction of Pre-B phonotactics based on morphological evidence.

8.6. Implications for historical morphosyntax

Although Pre-B derivation is largely lexical, its functions imply a system of referential delimitation (*-na*), a system of localization (-(V)r(a)), and a system of functional characterization (*-ri*). These categories anticipate—but do not coincide with—later syntactic categories in Basque. This suggests that Pre-B morphology contributed to the emergence of later morphosyntactic distinctions, even if indirectly.

8.7. A framework for future research

The minimal model proposed here provides a foundation for further work on Pre-B phonotactics, lexical reconstruction, internal stratification, morphosyntactic emergence, and the interface between derivation and deixis. Because the model is testable and reproducible, it can be refined as new data or analyses become available.

9. Conclusion

This study has identified and analyzed six derivational mechanisms that can be securely attributed to the Pre-B morphological layer. These mechanisms—localization (-(V)r/-(V)ra), collectivization (-ta/-eta), individuation (-na), light nominalization (a-/e-), agentive/instrumental derivation (-ri), and qualitative specification (s $\sim$ z)—form a coherent and internally structured system irreducible to either Proto-Basque or modern Basque morphology. Their recurrence, segmentability, and co-distribution with other Pre-B traits confirm that they belong to an earlier morphological stratum.

Beyond the identification of individual patterns, the analysis has demonstrated that Pre-B derivation exhibits layered organization, functional complementarity, and domain-specific constraints. These properties reveal a derivational architecture in which morphological operations combine in predictable sequences, shaping lexical bases into localized, collective, individuated, or functionally specified entities. This architecture aligns with the Pre-B deictic system reconstructed in Girard (2026a), suggesting that deixis and derivation formed an integrated morphological domain.

The minimal model proposed in Section 7 provides a conservative yet testable representation of this architecture. It captures the essential components of Pre-B derivation without over-interpreting the available data, and it offers a framework for future research on Pre-B phonotactics, lexical stratification, and the emergence of later morphosyntactic categories.

Taken together, these findings demonstrate that Pre-B was not a static lexical substrate but a morphologically active system with its own derivational grammar. The reconstruction of this system contributes to a more precise understanding of the internal structure of the Basque lexicon and provides a foundation for further inquiry into the earliest recoverable stages of the language.

References

Aldai, Gontzal. 2020. Complex predicates, simple inflecting verbs, and "uninflecting verbs" in Pre-Basque. *Linguistics* 58(6). 1609–1645.

Campbell, Lyle. 2013. *Historical linguistics: An introduction*. 3rd edn. Edinburgh: Edinburgh University Press.

Egurtzegi, Ander. 2014. *Towards a phonetically grounded diachronic phonology of Basque*. Vitoria-Gasteiz: University of the Basque Country (UPV/EHU) dissertation.

Faria, António Marques de. 2025. Crónica de onomástica paleo-hispânica (33). *Revista Portuguesa de Arqueologia* 28. 155–197.

Fox, Anthony. 1995. *Linguistic reconstruction: An introduction to theory and method*. Oxford: Oxford University Press.

Girard, Steve. 2026a. Pour une définition opératoire du « Pré-basque » : statut, périmètre, utilité. Québec: Zenodo. 1–9. https://doi.org/10.5281/zenodo.18795462.

Girard, Steve. 2026b. For an operational definition of "Pre-Basque": Status, scope, utility and reproducible protocols. Québec: Zenodo. 1–8. https://doi.org/10.5281/zenodo.18829186.

Girard, Steve. 2026c. Tests empirical and comparative for the hypothesis of an Eurasian palatal change in Basque. Québec: Zenodo. 1–10. https://doi.org/10.5281/zenodo.18856708.

Girard, Steve & K. 2026d. The Pre-Basque deictic system: A reconstruction. Manuscript submitted for publication. 1–39.

Gorrochategui, Joaquín. 1995. Los Pirineos entre Galia e Hispania: las lenguas. *Veleia* 12. 181–234.

Gorrochategui, Joaquín. 2009. Vasco antiguo: Algunas cuestiones de geografía e historia lingüísticas. *Palaeohispanica* 9. 539–555.

Gorrochategui, Joaquín & Joseba A. Lakarra. 1996. Nuevas aportaciones a la reconstrucción del Protovasco. In Francisco Villar & José d'Encarnação (eds.), *La Hispania prerromana: Actas del VI Coloquio sobre lenguas y culturas prerromanas de la Península Ibérica*, 101–145. Salamanca: Universidad de Salamanca.

Heine, Bernd & Tania Kuteva. 2002. *World lexicon of grammaticalization*. Cambridge: Cambridge University Press.

Lakarra, Joseba A. 1995. Reconstructing the root in Pre-Proto-Basque. In José Ignacio Hualde, Joseba A. Lakarra & R. L. Trask (eds.), *Towards a history of the Basque language*, 189–206. Amsterdam: John Benjamins.

Lakarra, Joseba A. 2005. Prolegómenos a la reconstrucción de segundo grado y al análisis del cambio tipológico en (proto)vasco. *Palaeohispanica* 5. 407–470.

Lakarra, Joseba A. 2006. Protovasco, munda y otros: Reconstrucción interna y tipología holística diacrónica. *Oihenart* 21. 229–322.

Lakarra, Joseba A. 2011. Erro monosilabikoaren teoria eta aitzineuskararen berreraiketa: Zenbait alderdi eta ondorio. *Fontes Linguae Vasconum* 113. 5–114.

Lakarra, Joseba A. 2013. Root structure and the reconstruction of Proto-Basque. In Mikel Martínez-Areta (ed.), *Basque and Proto-Basque: Language-internal and typological approaches to linguistic reconstruction*, 173–221. Frankfurt am Main: Peter Lang.

Lakarra, Joseba A. 2018. CVC-edge alternancies and the foundations of the analysis of Proto-Basque word families. *Anuario del Seminario de Filología Vasca "Julio de Urquijo" (ASJU)* 52(1/2). 389–440.

Martínez-Areta, Mikel (ed.). 2013. *Basque and Proto-Basque: Language-internal and typological approaches to linguistic reconstruction*. Frankfurt am Main: Peter Lang.

Michelena, Luis. 1977. *Fonética histórica vasca*. 2nd edn. San Sebastián: Diputación Foral de Guipúzcoa.

Santazilia, Eñaut. 2013. Noun morphology. In Mikel Martínez-Areta (ed.), *Basque and Proto-Basque: Language-internal and typological approaches to linguistic reconstruction*, 223–281. Frankfurt am Main: Peter Lang.

Trask, R. L. 1997. *The history of Basque*. London: Routledge.